

A Reliability and Calibration Benchmark of QSVM Against Strong Tabular Learners on NSL-KDD and UNSW-NB15

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Abstract—An intrusion detector is deployed on the strength of its probabilities, yet quantum-kernel classifiers are almost always evaluated on accuracy alone. We benchmark a QSVM-zz against Random Forest, XGBoost, an MLP and an SVM-RBF baseline under one zero-leakage pipeline and one recalibration protocol, on NSL-KDD and UNSW-NB15. The natural way to pose the question is unavailable: the rare-attack union consists entirely of attack records, so binned calibration error collapses there to one minus mean confidence — an identity we verify to 2.2×10^{-16} . Measured where both classes are present, over 10 paired runs with Holm-corrected Wilcoxon tests, the quantum kernel is better calibrated than both tree ensembles in every setting ($\Delta\text{ECE} = -0.0415$, $p_{\text{Holm}} = 0.0039$ on NSL-KDD; -0.0694 , $p_{\text{Holm}} = 0.0039$ on UNSW-NB15) and at their own most favourable representation, while an MLP is better calibrated everywhere. Post-hoc recalibration helps the quantum kernel and no other model, and lowering the decision threshold recovers the rare classes for the margin-based models alone. One measured property explains both: a forest emits a distinct probability for 2.3% of test records against 73.6% for the quantum kernel, and a monotone recalibrator cannot separate tied scores. Of 80 Holm-corrected comparisons 25 favour the quantum kernel and 30 favour a classical baseline: the advantage is confined to calibration, and it is absent from ranking.

Index Terms—Quantum machine learning, quantum kernel, calibration, intrusion detection, NSL-KDD, UNSW-NB15.

I. INTRODUCTION

An intrusion detection system is rarely deployed as a bare classifier. It is deployed behind a threshold, feeding a queue that a human or a downstream policy has to triage, and the quantity that governs that queue is a probability rather than a label. A detector that is right often but confident at the wrong moments is harder to operate than one that is slightly less accurate and honest about it. Accuracy therefore answers only half of the deployment question.

Quantum-kernel classifiers are almost always evaluated on the other half alone. Since quantum feature maps were proposed as a route to expressive kernels [3], [4], and since separations were established for carefully constructed learning problems [5], the intrusion-detection literature has measured them by accuracy and F_1 . On tabular data of this kind, gradient-boosted trees and random forests usually win that contest [23], which makes an accuracy-only comparison both

unfavourable to the quantum kernel and uninformative about deployment.

This paper asks the other half of the question: when one of these models reports a probability, how far can that number be trusted? We find first that the natural way to pose the question — restrict the estimate to the rare attack classes that matter most — is unavailable for a structural reason, and that the quantity such a restriction produces is not a calibration error at all. We then measure calibration where it is well-defined, on two corpora and at two circuit widths.

Contributions

- 1) A measurement result: on a subset drawn by attack category, every record carries the same label, so binned calibration error is identically one minus mean confidence and cannot detect miscalibration. We verify the identity to 2.2×10^{-16} across every model and run, and give the non-degenerate counterpart.
- 2) A calibration benchmark of a quantum kernel against four baselines under one pipeline and one recalibration protocol, on two corpora and at two circuit widths, over 10 paired runs with Holm-corrected Wilcoxon tests.
- 3) The finding that the quantum kernel is better calibrated than both tree ensembles in every setting tested, and is not better than an MLP.
- 4) The finding that post-hoc recalibration helps the quantum kernel in every run and under every recalibrator tested, degrades the MLP and both tree ensembles in every run, and leaves the classical RBF kernel statistically unchanged.
- 5) A mechanism for that finding, measured rather than conjectured: the tree ensembles' scores are quantised and saturated, a monotone recalibrator cannot separate tied records, and the same property explains why no decision threshold recovers the rare classes for them.

Relation to the submitted version

This paper is a substantially revised version of an earlier submission, and five of that version's claims do not survive the revised protocol. We state them here rather than in a footnote.

(i) *The headline rare-attack result is withdrawn.* That version reported $\text{ECE}_{\text{rare}} = 0.4503$ for the quantum kernel against 0.5387 for the RBF kernel and concluded it was the most trustworthy model on the rare classes. The estimate was computed on 10 rare test records, and Section III shows that on

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a subset of one class the quantity is not a calibration error at all. Both the measurement and its interpretation are replaced.

(ii) *The low-data claim is narrowed.* That version placed the quantum kernel first from $N \geq 200$ onward. Measured where calibration is well-defined it leads at $N = 99$ only, and the MLP leads at every larger budget (Table IV).

(iii) *The balanced-operating-point claim is withdrawn.* The best-calibrated model under a balanced prior is MLP, not the quantum kernel (Table V).

(iv) *The recalibration claim is corrected in both directions.* That version concluded Platt scaling suits margin-based quantum and SVM models. It helps the quantum kernel alone; on the classical SVM no recalibrator is distinguishable from leaving the score alone ($p_{\text{Holm}} \geq 0.6973$ for all three). The finding is narrower than claimed and, once isotonic and temperature scaling are added, more robust.

(v) *The identical-footing claim was false.* That version reused cached quantum models whose training subset differed from the subset used to fit their calibrator, while the tree baselines were fitted and calibrated on the same data. Every model here is trained from scratch on one subset.

All five were found by re-running the protocol at a scale the original design could not reach, and not in response to any external report.

Reproducibility

Every number in this paper is emitted by a script into an artifact file and read back into the text through a generated macro; no figure is drawn by hand and no value is typed into the prose. The accompanying audit recomputes each of them from the artifacts, checks that the text contains no literal number, and asserts the claims the paper is and is not entitled to make — including the negative ones. A separate script verifies the identity of Section III numerically across every model and run. The full pipeline, the artifacts and the checks are released with the paper.

II. BACKGROUND AND RELATED WORK

A. Quantum Kernel and the zz-FeatureMap

Definition 1 (Quantum kernel). *For a data-dependent unitary $U : \mathbb{R}^n \rightarrow \mathcal{U}(2^n)$ with state map $\phi(\mathbf{x}) = U(\mathbf{x})|0\rangle^{\otimes n}$, the induced quantum kernel is $K^Q(\mathbf{x}, \mathbf{x}') = |\langle 0|U^\dagger(\mathbf{x}')U(\mathbf{x})|0\rangle|^2$.*

We use the depth- r zz-FeatureMap with full entanglement, $U_{\text{zz}}^{(r)}(\mathbf{x})$, whose entangling layer encodes pairwise feature products as quantum phases. The embedding width is not tuned here: it is taken from the companion study’s three-stage selection rule [2], which screens candidate widths on retained variance, then on kernel–target alignment, then on two-qubit gate cost, and returns 4 qubits at $r=2$ on NSL-KDD and 6 on UNSW-NB15. Reusing it without modification is what lets the present paper isolate classifier reliability from embedding choice.

What such a kernel can buy is itself contested. Huang *et al.* [27] show that access to training data shifts the comparison substantially, and that a quantum model’s advantage must be argued against classical learners given the same data rather

than against a worst case. We take that as the standard the comparison here has to meet, and apply it to reliability rather than to accuracy.

B. Confidence Calibration

Definition 2 (Calibration). *A probabilistic classifier producing $\hat{p}(\mathbf{x}) \in [0, 1]$ is perfectly calibrated if $\Pr(y=1 \mid \hat{p}(\mathbf{x})=q) = q$ for all q .*

Expected Calibration Error (ECE) and the Brier score are the standard scalar summaries of the gap from perfect calibration [7], [8], and a reliability diagram is the graphical counterpart. The Brier score is a strictly proper scoring rule [17], so it is minimised only by the true conditional probability; ECE is not, and is an estimator with a free bin count whose bias is known to be substantial [20].

Which models miscalibrate, and in which direction, is not arbitrary. Niculescu-Mizil and Caruana [19] showed that maximum-margin methods push probability mass away from 0 and 1 while other families push it toward the extremes, and that the shape of the distortion determines which post-hoc correction can repair it. That observation is the classical ancestor of the mechanism we measure in Section VII. Modern high-capacity classifiers are systematically over-confident and the effect grows with capacity [6], motivating post-hoc recalibration such as Platt scaling [9], a logistic map from decision scores to probabilities, and isotonic regression [18], its non-parametric counterpart. Neither is guaranteed to help: Kumar *et al.* [20] report that scaling methods are less calibrated than their measured error suggests. Calibration is rarely reported in quantum-IDS studies, and never compared against strong tabular learners.

The gap is not an oversight so much as an inheritance. The applied quantum-IDS literature grew out of demonstrations that a quantum feature map can separate classes at all, and the natural way to report such a demonstration is accuracy against a classical baseline. Payares and Martínez-Santos [13] compare quantum and classical models on denial-of-service detection by accuracy; Suryotrisongko and Musashi [14] evaluate a hybrid quantum–classical architecture on botnet traffic on the same footing; Kalinin and Krundyshev [15] survey quantum techniques for intrusion detection without reporting a probability-quality measure. None of the three reports a calibration curve, a Brier score, or an ECE, and none compares against a gradient-boosted tree. Table I places them against the three gaps this paper addresses.

The literature has continued to grow along the same axis. Abreu *et al.* [28] report a hybrid quantum–classical IDS evaluated on public corpora in binary and multiclass settings, and Kukliansky *et al.* [29] evaluate quantum neural networks for network anomaly detection on noisy hardware rather than in simulation — the one axis this paper cannot reach, and which Section IX names as its own next experiment. We place these two outside Table I: scoring a work against the three axes requires reading what it measures, and we restrict the table to works we have assessed in full.

That matters more here than it would elsewhere, because the accuracy comparison is the one a four-qubit kernel is

least likely to win. On tabular data the tree ensembles are the reference point [11], [12], and a representation compressed to four components to fit a NISQ budget gives the quantum branch little room. Measuring only accuracy therefore both understates what a quantum kernel offers and leaves the deployment question — whether the reported probability can be acted on — entirely open.

The calibration literature supplies the other half of the setting. Guo *et al.* [6] established that modern high-capacity classifiers are systematically over-confident and that the effect worsens with capacity, which motivated the post-hoc correction we apply uniformly here. Two features of our results sit awkwardly with that picture and are worth flagging early: on this corpus the models are *under*-confident rather than over-confident, and the correction helps exactly one of the five. Section VII argues that both follow from a property of the score distributions rather than from model capacity.

Proposition 1 (Platt scaling preserves ranking). *Platt scaling applies a strictly increasing logistic transform $\sigma(As + B)$, $A > 0$, to a score s . Because the transform is monotone, the induced ordering of samples is unchanged; hence threshold-free ranking metrics (AUC-ROC, AUC-PR) are invariant to Platt scaling, while binning-based calibration metrics (ECE) and the Brier score are not.*

Proposition 1 is the methodological reason our comparison can apply one uniform Platt protocol to every model without distorting the ranking comparison: any AUC difference reflects the raw model ordering, while ECE/Brier differences reflect calibration.

C. Strong Tabular Baselines

Random Forests [11] and gradient-boosted trees (XGBoost [12]) are the strongest non-deep models on tabular data and the comparators a reviewer expects for any practical IDS claim; we adopt both alongside SVM-RBF [10]. This is not folklore: on a benchmark of tabular datasets drawn from varied domains, with hyper-parameter search accounted for, Grinsztajn *et al.* [23] found tree ensembles still ahead of deep architectures at the data scale studied here.

D. Position Against Prior Work

Table I contrasts representative QSVM-IDS works against the three gaps. None reports calibration, compares against strong tabular learners, or separates ranking from trust. The present paper is, to our knowledge, the first calibration-centric reliability study of a quantum kernel for IDS.

III. WHY THE RARE-ATTACK SUBSET CANNOT ANSWER THE QUESTION

The obvious way to ask whether a detector is trustworthy on the attacks that matter is to restrict the calibration estimate to those attacks. On both corpora this does not work, for a reason that is structural rather than statistical.

The rare union is defined by attack category — User-to-Root and Remote-to-Local on NSL-KDD, and Worms, Shellcode,

TABLE I
COVERAGE OF THE THREE GAPS BY REPRESENTATIVE QSVM-IDS WORKS (G1: STRONG TABULAR BASELINES; G2: CALIBRATION; G3: RANKING VS. TRUST). A CHECK DENOTES THE AXIS IS EXPLICITLY STUDIED.

Work	G1	G2	G3
Payares <i>et al.</i> [13]	✗	✗	✗
Suryotrisongko <i>et al.</i> [14]	✗	✗	✗
Kalinin <i>et al.</i> [15]	✗	✗	✗
Companion (Paper 1) [2]	✗	partial	✗
This work	✓	✓	✓

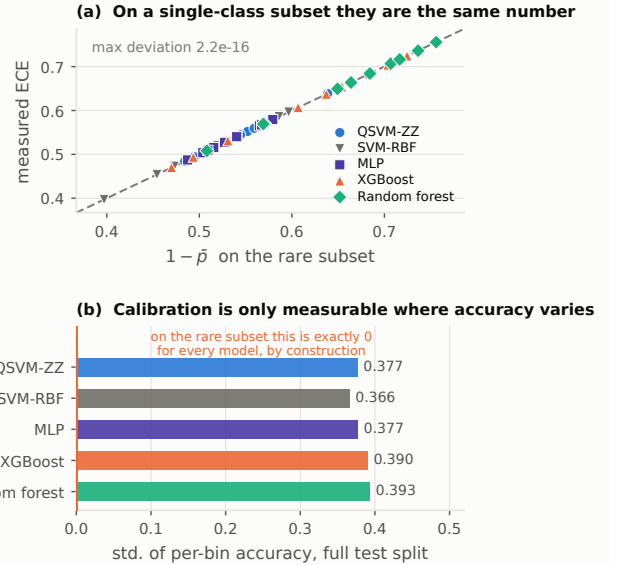


Fig. 1. (a) Measured calibration error on the rare subset against $1 - \bar{p}$, for every model and run on both corpora: they are the same number. (b) Calibration is only measurable where per-bin accuracy varies, which on the rare subset it cannot.

Backdoor and Analysis on UNSW-NB15. Every record in it is an attack, so its binary label is one. Under equal-frequency binning each bin B then contains only positives and $\text{acc}(B) = 1$ identically, so

$$\text{ECE} = \sum_B \frac{|B|}{n} |\text{acc}(B) - \text{conf}(B)| = \sum_B \frac{|B|}{n} (1 - \text{conf}(B)) = 1 - \bar{p}, \quad (1)$$

where $\bar{p}$ is the mean predicted attack probability on the subset. The right-hand side does not depend on the binning at all. The quantity is a measure of confidence on known attacks, close in spirit to recall; it is not a calibration error, and no model can be penalised by it for being over-confident.

We verify Eq. (1) numerically for every model and every run (Fig. 1(a)): the largest absolute deviation is 2.2×10^{-16} . We also verify the converse, that the estimate on the full test split is not degenerate: there both classes are present, per-bin accuracy varies, and the identity fails by a wide margin. All calibration numbers below are therefore measured on the full test split of each corpus, and we report the rare-subset quantity under its own name where it is informative.

Algorithm 1 Uniform probability and reliability protocol

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- 1: **input** train split (X_{tr}, y_{tr}) , test split (X_{te}, y_{te}) , model M , bins B
 - 2: fit M on (X_{tr}, y_{tr})
 - 3: $s_{tr} \leftarrow \text{score}(M, X_{tr})$, $s_{te} \leftarrow \text{score}(M, X_{te})$ {decision function, or log-odds for probabilistic models}
 - 4: fit recalibrator g on (s_{tr}, y_{tr}) {no test label is seen}
 - 5: $\hat{p} \leftarrow g(s_{te})$
 - 6: partition $\hat{p}$ into B equal-frequency bins
 - 7: **return** ECE, BS, AUC-PR from Eq. (2)
-

IV. RELIABILITY EVALUATION FRAMEWORK

A. What we measure

For a test set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and calibrated probabilities $\hat{p}_i$, partitioned into B equal-frequency bins,

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad \text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2, \quad (2)$$

with acc and conf the empirical accuracy and mean confidence in a bin. Equal-frequency rather than equal-width binning keeps every bin populated on a skewed score distribution. We report AUC-PR alongside, because ranking quality and probability quality are separate axes and a model can lead on one while trailing on the other.

B. One recalibration protocol for every model

Each model is reduced to a one-dimensional score before recalibration: the decision function for the kernel machines, the log-odds of the positive-class probability for the tree ensembles and the MLP. A single logistic map is then fitted on the *training* scores and applied to the test scores, so no test label enters the calibrator. By Proposition 1 this leaves AUC-PR and AUC-ROC untouched, which is what allows one protocol to be applied uniformly without distorting the ranking comparison.

Algorithm 1 states the procedure applied identically to every model. Two properties matter. The recalibrator sees only training scores and training labels, so no test information reaches the probability; and reducing each model to a one-dimensional score before recalibration is what makes a single protocol applicable to a kernel machine, a tree ensemble and a neural network at once. Where the literature usually compares calibration numbers produced by whatever each library happens to emit, every number here comes from the same seven lines.

C. The five questions

We ask the same question under five conditions, and the conditions are chosen because they correspond to things that change in deployment rather than to things that are convenient to vary.

- 1) **Operating calibration.** Calibration on the full test split of each corpus (Table II).

- 2) **Representation.** Whether the tree ensembles are disadvantaged by sharing the quantum kernel’s low-dimensional input (Fig. 4).
- 3) **Label scarcity.** Calibration against training-set size (Table IV).
- 4) **Class-prior shift.** Calibration when the deployment mix differs from the benchmark mix (Table V).
- 5) **Temporal drift.** Calibration on the harder KDDTest-21 split, whose records the original benchmark’s classical learners all misclassified.

Each is reported with paired tests and effect sizes rather than with point estimates, and we report the conditions in which the quantum kernel does not lead as prominently as those in which it does.

V. EXPERIMENTAL SETUP

All models share a zero-leakage front end: one-hot encoding, SELECTKBEST, PCA and a min-max map onto $[0, \pi]$, every transformer fitted on the training subset alone. Reliability is evaluated over 10 independent stratified training subsets of 1 000 records. Every model, quantum and classical, is trained from scratch on the same subset; the submitted version reused cached quantum models whose training subset differed from the one used to fit their calibrator, so the quantum and tree branches were not in fact on an identical footing.

A. Corpora

NSL-KDD [1] is the de-duplicated revision of KDD’99 and remains the reference corpus for this line of work. Its training split holds 125 973 records at 46.5% attack, its test split 22 544 at 56.9%, and attacks are grouped into 5 categories of which User-to-Root and Remote-to-Local are the rare and security-critical ones. The test split deliberately contains attack types absent from training, which is what makes it a test of generalisation rather than of memorisation — and, as Fig. 3(a) shows, what makes every model on it under-confident.

UNSW-NB15 [30] was built to address the age of KDD’99: it is synthesised from modern traffic with a contemporary attack mix. Its training split holds 175 341 records at 68.1% attack and its test split 82 332 at 55.1%, over 10 attack categories. The rare union here is Worms, Shellcode, Backdoor and Analysis. The two corpora differ in era, in attack taxonomy and in the direction of their prior shift between splits, which is what makes agreement across them informative.

B. Test splits

NSL-KDD is evaluated on the full KDDTest+ split, 22 544 records of which 2 952 belong to the rare union; UNSW-NB15 on its full test split, 82 332 records of which 1 682 are rare. The submitted version used a 100-record sample of KDDTest+ holding 10 rare records. The reason for the small sample was the cost of the quantum Gram matrix; evaluating the feature map in closed form removes that cost, so the constraint no longer binds.

C. Circuit width

The companion study's selection rule returns 4 qubits on NSL-KDD and 6 on UNSW-NB15, so comparing the corpora at their selected widths would change the dataset and the width together. We therefore run UNSW-NB15 at both widths and report both.

D. What the quantum branch costs

A recommendation is only actionable if the recommended model is affordable, and for a quantum kernel that is not a detail: the cost of the Gram matrix is precisely why the submitted version evaluated on 100 records rather than 22 544. Evaluating the feature map in closed form removes it. Averaged over five runs at $N = 1000$ training records and 22 544 test records, the quantum kernel fits in 0.020 s and scores the full test split in 0.325 s. The classical RBF kernel, which carries the same support-vector arithmetic at inference, takes 0.332 s on the same split — within a few percent. Fitting is the cheapest of the five models, because the Gram matrix reduces to one dense product; only the MLP, at 0.763 s, is an order of magnitude slower to train.

The cost objection to quantum kernels is therefore not operative at this scale. It returns at larger N , where the Gram grows quadratically, and on real hardware, where each entry needs repeated circuit execution rather than one matrix product. Both are outside what we measure.

E. Statistics

We report paired Wilcoxon signed-rank tests with bootstrap confidence intervals and d_z , following the standard recommendation for comparing classifiers over repeated runs [22], and correct for multiplicity within baseline families by the Holm sequentially rejective procedure [21]. Ten runs are the minimum for which this is meaningful: with five paired runs the smallest attainable two-sided p is 0.0625, so no comparison can reach significance whatever the effect size.

VI. RESULTS

A. The census

Before the individual comparisons, the tally. Across every setting, metric and baseline, the protocol produces 80 Holm-corrected comparisons: 25 favour the quantum kernel, 30 favour a classical baseline, and 25 are inconclusive. There is no general advantage here, and a paper that reported only its wins could have drawn a very different picture from the same runs.

The tally covers three families of metric and nothing else. On calibration measured where both classes are present — ECE and the Brier score on the full test split — the quantum kernel takes 16 of 32. On ranking — AUC-PR and F_1 — it takes 0 of 32. On the Brier score restricted to the rare union it takes 9 of 16. The three families sum to 80, so no comparison is counted twice and none is left out of the report.

One quantity is deliberately absent. ECE on the rare union is excluded from the tally, because Section III shows it equals $1 - \bar{p}$ identically there and is not a calibration error; counting it

among calibration wins would contradict that section. It would have added 16 comparisons. It is still reported, under its own name and per category, in Section VI. The Brier score on the same subset is kept, because a proper scoring rule does not degenerate when the subset carries one label.

The structure inside the tally is the result. The quantum kernel does not rank better than any baseline anywhere, and it calibrates better than half of them everywhere. That is the shape of the claim this paper makes, and the reason a single scalar comparison would have hidden it.

B. Calibration against the tree ensembles

The quantum kernel is better calibrated than both tree ensembles in every setting tested. On NSL-KDD the paired difference is -0.0415 $[-0.0565, -0.0268]$ ($p_{\text{Holm}} = 0.0039$, $d_z = -1.64$) against Random Forest and -0.0337 $[-0.0492, -0.0189]$ ($p_{\text{Holm}} = 0.0039$) against XGBoost. On UNSW-NB15 at the selected width the same comparisons give -0.0694 ($p_{\text{Holm}} = 0.0039$, $d_z = -4.76$) and -0.0670 ($p_{\text{Holm}} = 0.0039$), and at 4 qubits -0.0859 and -0.0667 . Eight of eight comparisons are corrected-significant in the quantum kernel's favour, and the effect is far larger on UNSW-NB15 than on NSL-KDD.

C. What the calibration curves look like

Fig. 3(a) plots the curve that Eq. (2) summarises. The shape is the same for all five models and it is not the one the calibration literature leads us to expect. Tree ensembles are usually reported as over-confident [6]; here every model is *under*-confident, lying above the diagonal in 7 of 10 bins for the quantum kernel and similarly for the rest. The observed attack fraction exceeds the predicted probability across most of the range.

The reason is the corpus rather than the models. KDDTest+ contains attack types absent from the training split, so records that a model scores low on familiarity grounds are nonetheless attacks. A detector tuned on this benchmark will under-alert if its probabilities are taken at face value, and that applies to the quantum kernel as much as to the baselines.

D. Which attacks the probabilities can be trusted on

Section III showed that a calibration estimate restricted to one attack category is exactly one minus the mean confidence on it. That quantity is still worth reporting — under its own name, and per category rather than pooled. Table III and Fig. 3(b) give it.

The ordering differs sharply by category. On Probe the quantum kernel is the weakest of the five (0.7184 against 0.9228 for XGBoost), and on Normal it is neither the most nor the least conservative. On the two rare categories the ordering reverses: the quantum kernel assigns the highest mean attack probability to (0.4483 against 0.3137 for Random Forest), and on R2L it is second behind the RBF kernel. This is the defensible form of the rare-attack claim: not that the quantum kernel is better calibrated there — that statement is not measurable — but that it is the least reluctant of the five to call a rare attack an attack.

TABLE II

CALIBRATION ERROR ON THE FULL TEST SPLIT OF EACH DATASET, MEAN $\pm$ STD OVER 10 RUNS. LOWER IS BETTER; BEST PER COLUMN IN BOLD. THE QUANTUM KERNEL IS BETTER CALIBRATED THAN BOTH TREE ENSEMBLES IN EVERY SETTING, AND IS NOT BETTER THAN THE MLP.

Model	NSL-KDD KDDTest+	NSL-KDD KDDTest-21	UNSW-NB15 (4 qubits)	UNSW-NB15 (6 qubits)
QSVM- $\mathbb{Z}$	0.1421 $\pm$ 0.0114	0.2935 $\pm$ 0.0267	0.1449 $\pm$ 0.0093	0.1531 $\pm$ 0.0149
SVM-RBF	0.1272 $\pm$ 0.0154	0.2872 $\pm$ 0.0249	0.1518 $\pm$ 0.0189	0.1474 $\pm$ 0.0171
MLP	0.1172$\pm$0.0114	0.2503$\pm$0.0187	0.1198$\pm$0.0073	0.1226$\pm$0.0096
XGBoost	0.1758 $\pm$ 0.0277	0.3351 $\pm$ 0.0471	0.2117 $\pm$ 0.0071	0.2201 $\pm$ 0.0121
Random forest	0.1836 $\pm$ 0.0303	0.3487 $\pm$ 0.0473	0.2308 $\pm$ 0.0083	0.2225 $\pm$ 0.0085

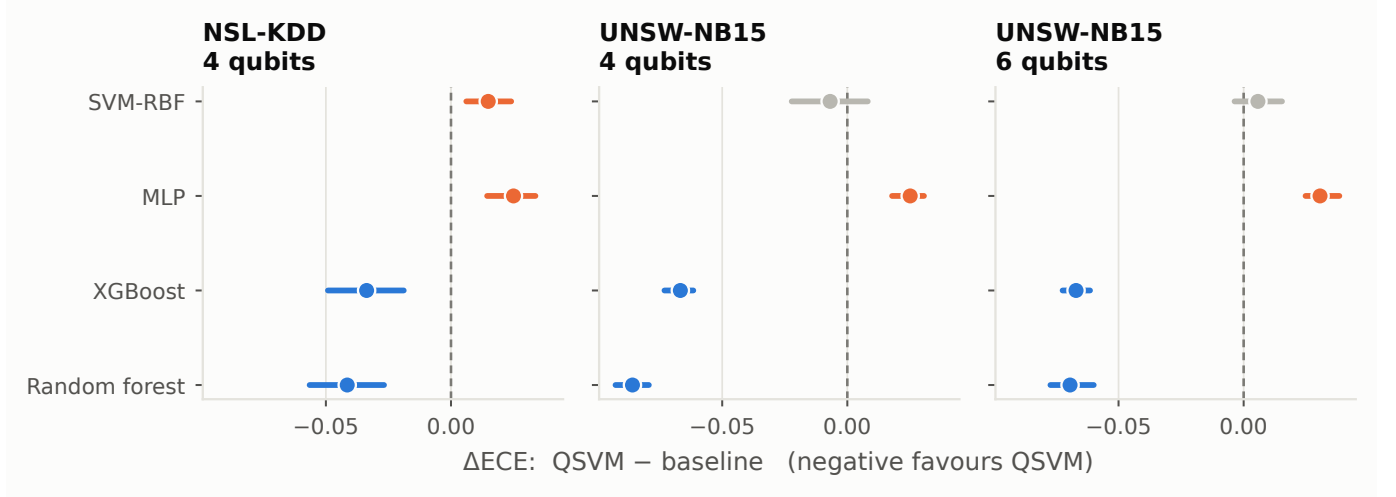


Fig. 2. Paired differences in calibration error, quantum kernel minus baseline, with bootstrap confidence intervals and Holm-corrected p . Blue favours the quantum kernel, orange the baseline, grey is inconclusive.

One observation applies to every model and matters more than the ordering. The mean probability on both rare categories falls below the decision threshold for all five: at a default cut-off, none of these detectors would raise an alert on a typical R2L or U2R record. The families differ in how far below they sit, not in whether they clear it.

E. Where the quantum kernel does not lead

The MLP is better calibrated than the quantum kernel in every setting (+0.0250 on NSL-KDD, $p_{\text{Holm}} = 0.0059$; +0.0305 on UNSW-NB15, $p_{\text{Holm}} = 0.0020$), and we report this as prominently as the wins. Against the classical RBF kernel the outcome is dataset-dependent: on NSL-KDD the RBF kernel is ahead (+0.0149, $p_{\text{Holm}} = 0.0195$), while on UNSW-NB15 the comparison is inconclusive at both widths ($p_{\text{Holm}} = 0.2754$ at 6 qubits). The claim this paper supports is a claim about tree ensembles, not a claim that a quantum kernel is the most reliable model available.

F. Are the tree baselines handicapped?

The trees share the quantum kernel’s low-dimensional representation, which invites the objection that they are compared in a form that suits them badly. We repeat them on the 20 selected features and on the full 122 one-hot encoding, holding every hyper-parameter fixed so that only the representation changes.

More features do not make the trees better calibrated. Random Forest’s calibration error moves from 0.1836 to 0.1858

and XGBoost’s from 0.1758 to 0.1818 — slightly worse in both cases, and in neither case enough to approach the quantum kernel. Ranking moves the other way and by a larger margin: Random Forest’s AUC-PR improves from 0.9481 to 0.9648. The point is not that the richer representation harms the trees much; it is that it does not help their calibration at all, so the comparison in Table II is made in the trees’ most favourable calibration setting rather than a cramped one.

G. When post-hoc recalibration helps

Platt scaling reduces the quantum kernel’s calibration error from 0.1969 to 0.1421 (+0.0547) and is the only model of the five it helps: Random Forest moves by -0.0531 and XGBoost by -0.0366 , both worse, and the classical SVM by -0.0102 .

This is not an artefact of choosing a logistic map. Isotonic regression and temperature scaling give the same picture: on the quantum kernel all three land within a narrow band (0.1421, 0.1398, 0.1399 against 0.1969 uncalibrated), while on the MLP and both tree ensembles all three are worse than leaving the native probability alone (Random Forest 0.1305 uncalibrated against 0.1787 isotonic).

Which of those movements are real is a separate question from their size, and the last column of Table VI answers it with the same paired test used everywhere else, Holm-corrected over the three recalibrators. Three verdicts come back, not two. All three maps improve the quantum kernel in 10/10 runs ($p_{\text{Holm}} = 0.0059$ for each), the largest mean gain being

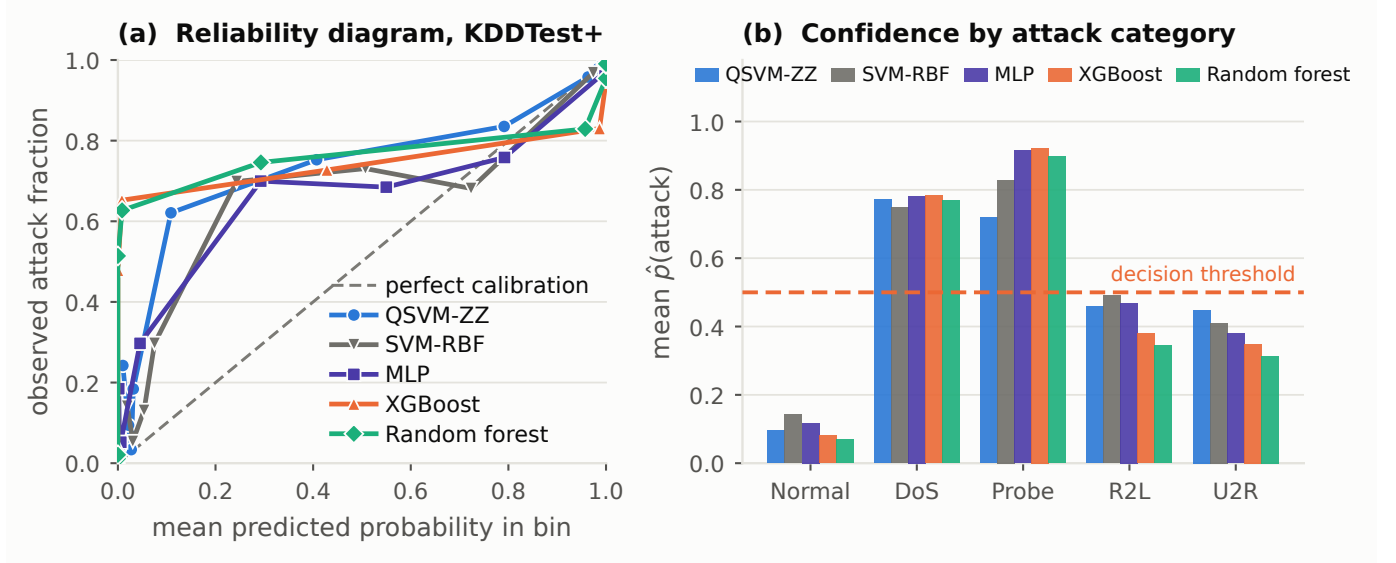


Fig. 3. (a) Reliability diagram on KDDTest+. Every model lies above the diagonal over most of the range: all five are under-confident rather than over-confident on this corpus. (b) Mean predicted attack probability by category. Both rare categories sit below the decision threshold for every model.

TABLE III

MEAN PREDICTED ATTACK PROBABILITY BY CATEGORY ON KDDTest+, OVER 10 RUNS. NORMAL IS THE NEGATIVE CLASS, SO LOWER IS BETTER THERE AND HIGHER IS BETTER ELSEWHERE. EVERY MODEL FALLS BELOW THE 0.5 DECISION THRESHOLD ON BOTH RARE CATEGORIES.

Category	n	QSVM-zz	SVM-RBF	MLP	XGBoost	Random forest
Normal	9 711	0.0960	0.1419	0.1180	0.0803	0.0693
DoS	7 460	0.7720	0.7491	0.7818	0.7830	0.7697
Probe	2 421	0.7184	0.8284	0.9159	0.9228	0.8975
R2L	2 885	0.4589	0.4902	0.4685	0.3797	0.3446
U2R	67	0.4483	0.4098	0.3787	0.3487	0.3137

+0.0571. All three make the MLP, XGBoost and Random Forest worse in every run — none of the three improves any of them even once, at the same corrected p . The RBF kernel is neither case: its best map, temperature scaling, moves the mean by +0.0031 but wins in only 5/10 runs ($p_{\text{Holm}} = 1.0000$), and its worst moves it by -0.0102 at $p_{\text{Holm}} = 0.6973$. We therefore claim that recalibration helps the quantum kernel and no other model, and *not* that it degrades every other model — for the RBF kernel the data do not separate the effect from zero in either direction. The distinction is not bookkeeping: the mechanism of Section VII predicts a null there rather than a loss, because the RBF kernel’s scores are as unsaturated as the quantum kernel’s (73.6% of test records receive a distinct probability against 73.6%) while its uncalibrated error is already the second lowest of the five, leaving a monotone map little room to help and little to spoil.

A global improvement can hide a local regression, so we repeat the comparison on the rare categories alone. It does not: the quantum kernel’s mean confidence there is 0.4536 under Platt, 0.4668 under isotonic and 0.4591 under temperature scaling, the highest of the five under every recalibrator, and no recalibrator pushes the tree ensembles above 0.3477. Temperature scaling is the only combination that lifts any model over the decision threshold on the rare union, and it does so for the RBF kernel (0.5059) rather than for the quantum kernel.

TABLE IV

CALIBRATION ERROR AGAINST TRAINING-SET SIZE ON KDDTest+, MEAN OVER 10 RUNS. BEST PER COLUMN IN BOLD. THE QUANTUM KERNEL LEADS AT THE SMALLEST SIZE ONLY.

Model	$N = 99$	$N = 200$	$N = 500$	$N = 1000$
QSVM-zz	0.1549	0.1402	0.1357	0.1421
SVM-RBF	0.1677	0.1361	0.1353	0.1272
MLP	0.1578	0.1299	0.1176	0.1172
XGBoost	0.1696	0.1625	0.1627	0.1758
Random forest	0.1752	0.1745	0.1765	0.1836

TABLE V

CALIBRATION ERROR UNDER CLASS-PRIOR SHIFT, MEAN OVER 10 RUNS. EACH CONDITION IS REBUILT FROM THE FULL TEST SPLIT. BEST PER COLUMN IN BOLD.

Model	Balanced	Attack-heavy	DoS-only
QSVM-zz	0.1250	0.1898	0.0746
SVM-RBF	0.1089	0.1718	0.0943
MLP	0.1016	0.1563	0.0933
XGBoost	0.1514	0.2031	0.1049
Random forest	0.1599	0.2136	0.1111

H. Training-set size

Reliability is not uniform in the amount of training data. Table IV sweeps $N \in \{99, \dots, 1000\}$, refitting every model on each subset. The best-calibrated model at the smallest size is QSVM-zz (0.1549 against 0.1578 for the MLP and 0.1752

TABLE VI

CALIBRATION ERROR BEFORE AND AFTER POST-HOC RECALIBRATION ON KDDTest+, MEAN OVER 10 RUNS. A CELL IS BOLD WHEN ITS MEAN IMPROVES ON LEAVING THE NATIVE PROBABILITY ALONE. THE LAST COLUMN IS THE PAIRED WILCOXON VERDICT, HOLM-CORRECTED OVER THE THREE RECALIBRATORS: IT IS WHAT THE TEXT CLAIMS, BECAUSE A MEAN CAN MOVE WITHOUT THE PER-RUN COMPARISON SUPPORTING IT.

Model	none	Platt	isotonic	temperature	Paired test (runs improved)
QSVM-zz	0.1969	0.1421	0.1398	0.1399	helps, 10/10
SVM-RBF	0.1171	0.1272	0.1254	0.1140	no effect, 3–5/10
MLP	0.1084	0.1172	0.1356	0.1144	hurts, 0/10
XGBoost	0.1392	0.1758	0.1802	0.1712	hurts, 0/10
Random forest	0.1305	0.1836	0.1787	0.1795	hurts, 0/10

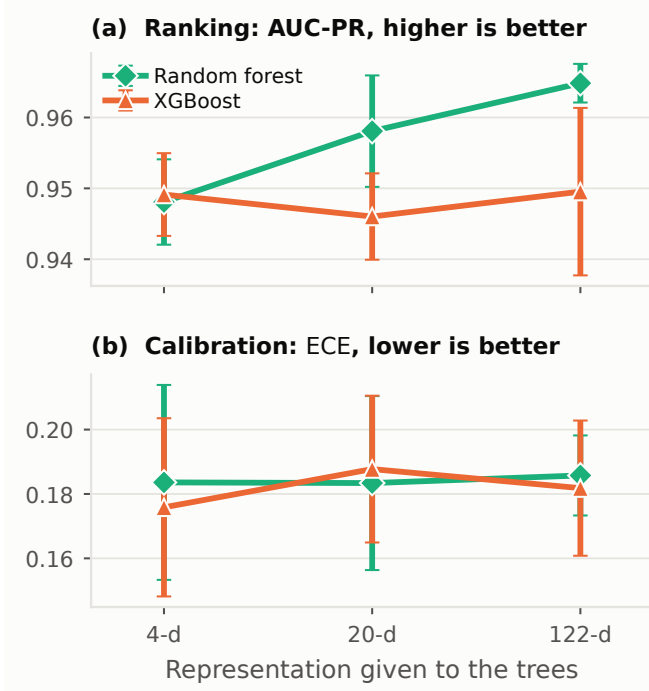


Fig. 4. Giving the tree ensembles a richer representation, with every hyper-parameter held fixed. Ranking improves; calibration gets worse.

for Random Forest) — the one condition in this study where the quantum kernel leads the field outright. Its relative standing falls as N grows: at $N = 1000$ it is 0.1421 against 0.1172 for the MLP. The advantage, such as it is, belongs to the low-data regime, which is also the regime in which a four-qubit kernel is cheapest to run.

I. Class-prior shift

A benchmark's class mix is not the deployment mix, and a calibrator fitted under one can fail under the other. We rebuild three operating conditions from the full test split rather than from a fixed sample (Table V): a balanced mix (19422 records), an attack-heavy mix (18332), and a DoS-only mix (17171). The best-calibrated model is MLP on the balanced mix (0.1016 against 0.1250 for the quantum kernel), MLP on the attack-heavy mix, and QSVM-zz on the DoS-only mix (0.0746). The quantum kernel is ahead of both tree ensembles under all three conditions (0.1599, 0.2136, 0.1111 for Random

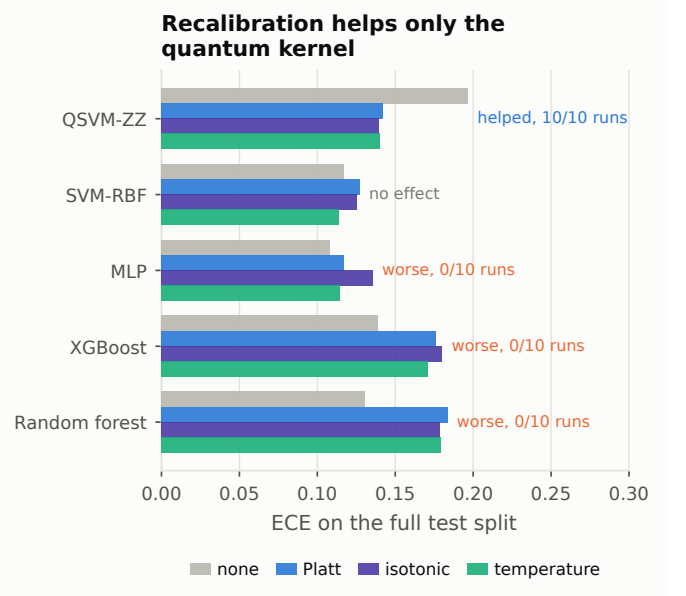


Fig. 5. Post-hoc recalibration helps the quantum kernel in all 10 runs, degrades the MLP and both tree ensembles in all of them, and leaves the RBF kernel unchanged. The labels are the Holm-corrected paired verdicts of Table VI rather than a comparison of means: the RBF kernel's mean does improve under temperature scaling, but it wins in only 5/10 runs.

Forest), which is the same ordering as Table II and not a separate effect.

J. Temporal drift

KDDTest-21 retains the records of KDDTest+ that the original benchmark's classical learners all misclassified, and is the standard stand-in for drift on this corpus. The ordering is unchanged there: the quantum kernel is better calibrated than Random Forest by -0.0553 $[-0.0782, -0.0345]$ ($p_{\text{Holm}} = 0.0039$, $d_z = -1.47$) and than XGBoost by -0.0416 ($p_{\text{Holm}} = 0.0273$), behind the MLP by $+0.0432$ ($p_{\text{Holm}} = 0.0059$), and inconclusive against the RBF kernel ($p_{\text{Holm}} = 0.6953$). Counting KDDTest-21 alongside the four settings of Table II, the quantum kernel beats both tree ensembles in ten of ten corrected comparisons.

K. Ranking is not reliability, as a decomposition

So far the separation of the two axes has been an observation about two columns of a table. The Murphy decomposition [16]

TABLE VII

MURPHY DECOMPOSITION OF THE BRIER SCORE ON KDDTEST+, BS = REL - RES + UNC, MEAN OVER 10 RUNS. LOWER REL IS BETTER CALIBRATED, HIGHER RES DISCRIMINATES BETTER, UNC DEPENDS ONLY ON THE DATA. THE TREE ENSEMBLES HOLD THE BEST RESOLUTION AND THE WORST RELIABILITY.

Model	REL ↓	RES ↑	UNC	BS
QSVM-zz	0.0514	0.1420	0.2452	0.1546
SVM-RBF	0.0392	0.1341	0.2452	0.1503
MLP	0.0355	0.1425	0.2452	0.1382
XGBoost	0.0848	0.1524	0.2452	0.1776
Random forest	0.0948	0.1548	0.2452	0.1852

TABLE VIII

OPERATING POINT. τ^* MAXIMISES MACRO F_1 ; U2R RECALL IS GIVEN AT THE DEFAULT THRESHOLD AND AT τ^* , WITH THE FALSE-POSITIVE RATE ON NORMAL TRAFFIC THAT τ^* COSTS. LOWERING THE THRESHOLD RECOVERS THE RARE CLASS FOR THE MARGIN-BASED MODELS AND NOT FOR THE TREE ENSEMBLES.

Model	τ^*	F_1^*	U2R @ 0.5	U2R @ τ^*	FPR @ τ^*
QSVM-zz	0.05	0.8569	0.422	0.825	0.185
SVM-RBF	0.10	0.8423	0.378	0.828	0.228
MLP	0.05	0.8528	0.300	0.940	0.236
XGBoost	0.05	0.8186	0.346	0.430	0.103
Random forest	0.05	0.8155	0.321	0.391	0.093

turns it into an identity. Writing BS = REL - RES + UNC over the same bins Eq. (2) uses, the reliability term REL is the squared calibration gap, the resolution term RES measures how far a model's bins depart from the base rate — that is, how much it discriminates — and UNC depends only on the data and is therefore identical across models.

Table VII and Fig. 6(b) give the result, and it is the cleanest statement of this paper's thesis. The best resolution belongs to Random forest (0.1548), and so does the *worst* reliability (0.0948). The best reliability belongs to MLP (0.0355) on a resolution that is lower (0.1425). The quantum kernel's resolution, 0.1420, is within a thousandth of the MLP's, while its reliability is 0.0514: it discriminates as well and calibrates less well. A single Brier number would have hidden all of this, because the two terms move in opposite directions and partly cancel.

L. Choosing an operating point

Section VI left an open question: if the mean probability on the rare categories is below the threshold for every model, what threshold would be needed, and what does it cost? Table VIII and Fig. 6(a) answer it, and the answer separates the model families more sharply than any calibration number does.

For the margin-based models, lowering the threshold works. The quantum kernel's U2R recall rises from 0.422 at the default to 0.825 at its macro- F_1 optimum $\tau^* = 0.05$, at a false-positive rate on normal traffic of 0.185. The MLP reaches 0.940 and the RBF kernel 0.828 on the same sweep.

For the tree ensembles it does not. Random Forest moves only from 0.321 to 0.391, and XGBoost from 0.346 to 0.430; across the entire sweep neither exceeds 0.391 and 0.430 respectively. Their probability mass on U2R is concentrated so near zero that no threshold recovers the class. This is a

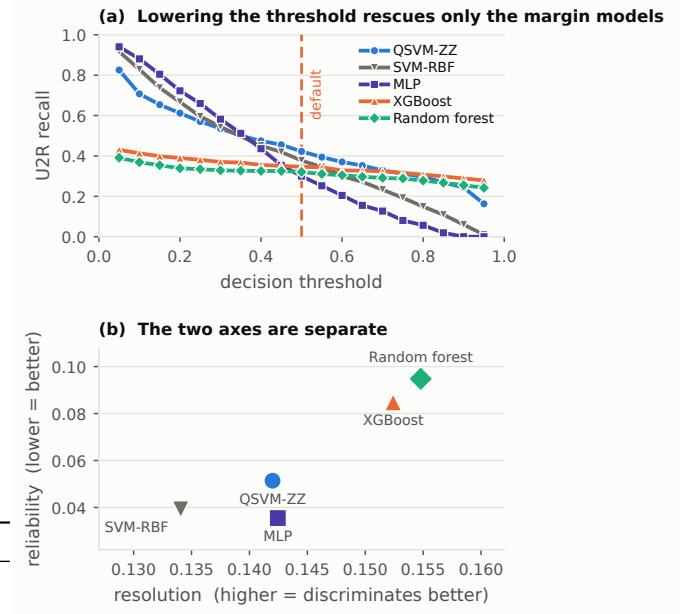


Fig. 6. (a) U2R recall against the decision threshold. The margin-based models recover the rare class as the threshold falls; the tree ensembles do not. (b) The Murphy decomposition separates the two axes: the tree ensembles sit top-right, discriminating best and calibrating worst.

deployment consequence of the calibration result rather than a restatement of it: a practitioner who responds to missed rare attacks by lowering the threshold will be rewarded with a margin-based detector and will not be with a tree ensemble.

VII. WHY THE FAMILIES DIFFER

Three results so far point the same way: recalibration helps the quantum kernel and not the tree ensembles; lowering the threshold recovers the rare class for the margin-based models and not for the trees; and the trees hold the best resolution together with the worst reliability. A single property of the score distributions accounts for all three, and it is measurable rather than conjectural.

A post-hoc recalibrator — Platt, isotonic or temperature — is a *monotone* map. It can move scores along the line but it cannot separate two records that received the same score. Whatever resolution the raw scores lack, no such map can supply. So the question is how much distinct information the raw scores carry before recalibration begins.

They differ by more than an order of magnitude. Over 10 runs on KDDTest+, the quantum kernel emits a distinct probability for 73.6% of test records and the RBF kernel for 73.6%, while Random Forest emits one for 2.3% and XGBoost for 19.4%. A forest's output is a vote fraction over a finite ensemble, and on a training set of this size most leaves are pure, so the fraction collapses onto a handful of values at the extremes: 53.8% of Random Forest's raw probabilities fall below 0.01 or above 0.99, against 0.0% for the quantum kernel, whose decision function is continuous and never saturates.

This explains the three results together. Ties and saturation give the trees high resolution, because the extremes separate

the classes sharply — and poor reliability, because the extremes are more confident than the data warrant. Recalibration cannot repair it, since the map is monotone and the tied records stay tied; indeed it makes matters worse, pushing 89.2% of Random Forest’s calibrated probabilities into the saturated tails. And no threshold recovers the rare classes for a model whose mass on them is already concentrated at zero.

The corollary is not about quantum computing. Any classifier whose scores are continuous and unsaturated will recalibrate well on this task, and any classifier whose scores are quantised and saturated will not. The quantum kernel is on the favourable side of that divide because it is a margin machine, not because it is quantum. What the quantum kernel adds over the classical margin machines on this evidence is not calibration quality but the resolution it achieves at that quality — 0.1420 against 0.1341 for the RBF kernel, at four qubits.

A. Where the account is incomplete

Saturation explains the contrast between the kernel machines and the tree ensembles. It does not explain the model that actually wins. The MLP saturates 48.2% of its raw outputs — much closer to the trees than to the kernels — and emits distinct values for 58.7% of records, again intermediate. On the saturation account it should calibrate somewhere between the two families. It does not: it has the lowest reliability term of the five, 0.0355, against 0.0514 for the quantum kernel and 0.0948 for Random Forest.

So saturation is necessary for the failure mode and not sufficient for the success. A plausible reading is that what matters is not saturation alone but whether the saturated mass is placed correctly: a network trained by cross-entropy is optimising a proper scoring rule directly and can be confident *and* right, whereas a forest’s vote fraction saturates as a by-product of leaf purity and carries no such guarantee. We cannot settle that from the evidence here — it would need an ablation over training objectives, which is a different experiment — and we flag it rather than present the mechanism as more complete than it is.

That the strongest baseline in this study is a small multilayer perceptron rather than a tree ensemble is itself worth stating plainly, because the quantum-IDS literature rarely includes one. On the deployment question this paper asks, the model to beat is not the one the accuracy benchmarks suggest.

VIII. RELIABILITY REGIME MAP

Table IX collects every condition in one place: which model is best calibrated, and whether the quantum kernel is better calibrated than both tree ensembles. Read down the last column, the answer is uniform — the quantum kernel beats both tree ensembles in every condition tested, across two corpora, two circuit widths, a drift split, four training budgets and three deployment priors. Read down the middle column, the answer is almost as uniform in the other direction: the MLP is the best-calibrated model nearly everywhere.

The two conditions where the quantum kernel leads outright are the smallest training budget and the DoS-only prior. Both are narrow, and we do not claim they generalise beyond what

TABLE IX

RELIABILITY REGIME MAP. FOR EACH CONDITION, THE BEST-CALIBRATED MODEL AND WHETHER THE QUANTUM KERNEL IS BETTER CALIBRATED THAN BOTH TREE ENSEMBLES. IT LEADS OUTRIGHT ONLY UNDER THE SMALLEST TRAINING BUDGET, AND BEATS BOTH TREE ENSEMBLES EVERYWHERE.

Condition	Best calibrated	Beats both trees
NSL-KDD, operating mix	MLP	✓
NSL-KDD, temporal drift	MLP	✓
UNSW-NB15, 4 qubits	MLP	✓
UNSW-NB15, 6 qubits	MLP	✓
Label scarcity, $N = 99$	QSVM-zz	✓
Label scarcity, $N = 200$	MLP	✓
Label scarcity, $N = 500$	MLP	✓
Label scarcity, $N = 1000$	MLP	✓
Balanced prior	MLP	✓
Attack-heavy prior	MLP	✓
DoS-only prior	QSVM-zz	✓

is in the table. But they are not arbitrary: the low-data regime is where a four-qubit kernel is cheapest to evaluate, and it is the regime in which the classical baselines have least to work with.

A decision rule

For a practitioner the map reduces to three statements. If the deployment mix and training budget resemble the benchmark, a small neural network is the better-calibrated choice and a quantum kernel buys nothing. If the candidate set is restricted to kernel machines and tree ensembles — which is the usual comparison in this literature — the quantum kernel is the better-calibrated choice, consistently and by a margin that survives correction. And in either case the probabilities on the rare categories should not be used at a default threshold without the recalibration of Section VI.

IX. THREATS TO VALIDITY

A. Construct validity

Binned calibration error is an estimator with a free parameter, and a different bin count gives a different number. This is not a detail we raise defensively: Kumar *et al.* [20] show the bias of the binned estimator is large enough that recalibrated models are routinely less calibrated than their reported ECE implies. We therefore use equal-frequency bins throughout and report the Brier score alongside every ECE, which needs no binning and is strictly proper [17]; the two agree on the ordering in Table II.

The identity of Section III is a stronger warning about the same family of estimators: a scalar summary can be well-defined arithmetically and still measure nothing, and the only defence is to check what varies in the subset one has restricted to. That failure mode is not peculiar to calibration metrics. Chicco and Jurman [24] show that F_1 and accuracy return inflated values on imbalanced data precisely because neither reads all four cells of the confusion matrix; Christen *et al.* [25] collect the same criticism of the F-measure and its alternatives; and in security specifically, Arp *et al.* [26] list inappropriate performance measures and the base rate fallacy among ten recurring pitfalls. Section III is an instance of that family in which the degeneracy is exact rather than statistical.

B. Internal validity

The recalibrators are fitted on the training split. For a model that can memorise that split — the tree ensembles above all — the fitted map is in-sample, which flatters it. This biases the comparison *towards* the trees, and they lose anyway, so the conclusion is safe in direction; but the magnitudes in Table VI would change with a held-out calibration split, and the finding that recalibration helps the quantum kernel alone should be read with that caveat. A second exposure is the hyper-parameters, which are inherited from the companion study rather than re-tuned here; re-tuning per model could move any row of Table II, and we did not do it because tuning each family on its own criterion would reintroduce exactly the asymmetry that Section V removes.

C. External validity

Two corpora, both offline benchmark captures with known idiosyncrasies, and both with attack types in the test split that are absent from training — which is why every model is under-confident in Fig. 3(a). Whether the ordering holds on live traffic is not established here. The quantum kernel is evaluated by exact statevector simulation, so shot noise and device noise are absent. The companion study reports that finite-shot estimation leaves macro F_1 close to the statevector value, but reports it from a single run and does not claim more; whether shot or device noise perturbs *calibration* is untouched by that check and by ours. It is the obvious next experiment, and on the mechanism of Section VII it is not a neutral one: noise would blur the decision scores, which is the direction that helps a recalibrator rather than hinders it. Evaluation on noisy hardware is already feasible for the adjacent task — Kukliansky *et al.* [29] run quantum neural networks for network anomaly detection on real devices — so the obstacle is access rather than method.

D. Statistical conclusion validity

Ten paired runs is the minimum at which a two-sided Wilcoxon test can reach $\alpha = 0.05$, so the design has no power to spare. Comparisons we report as inconclusive — the quantum kernel against the RBF kernel on three of four settings — may be real effects this design cannot resolve, and should not be read as evidence of equivalence. The comparisons we do claim clear Holm correction within their baseline family, but the families were fixed in advance rather than chosen after seeing the results.

X. CONCLUSION

We set out to ask whether a quantum-kernel intrusion detector can be trusted at the level of its probabilities, and found first that the question has to be posed carefully. Restricting a binned calibration estimate to the rare attack classes yields a quantity that is identically one minus mean confidence, because every record in that subset carries the same label; it measures confidence on known attacks and cannot detect over-confidence at all. Reported as a calibration error it would overstate what has been measured.

Measured where both classes are present, the result is narrow but it holds everywhere we looked. Across two corpora and two circuit widths, over 10 paired runs with Holm-corrected tests, the quantum kernel is better calibrated than both tree ensembles in all eight comparisons, and the margin is several times larger on UNSW-NB15 than on NSL-KDD. It is not better calibrated than a small multilayer perceptron anywhere, and against a classical RBF kernel the outcome depends on the corpus. Giving the trees the full one-hot representation improves their ranking without improving their calibration, so the comparison is made on terms that favour them. Post-hoc recalibration helps the quantum kernel and no other model, whether the recalibrator is Platt, isotonic or temperature scaling.

Underneath the benchmark there is a mechanism, and it is not a quantum one. A post-hoc recalibrator is a monotone map and cannot separate records that share a score. The tree ensembles emit a distinct probability for only 2.3% of test records against 73.6% for the quantum kernel, and saturate 53.8% of their outputs at the extremes against 0.0%. That single property accounts for all three of our secondary findings at once: why the trees hold the best resolution and the worst reliability, why no recalibrator repairs them, and why lowering the threshold recovers the rare classes for the margin-based models alone. The quantum kernel lands on the favourable side of that divide because it is a margin machine, and what it adds over the classical margin machines is resolution at a comparable calibration, not calibration itself.

The practical reading is modest. A four-qubit kernel is not the most reliable model available for this task, but among the families an intrusion-detection practitioner would actually reach for first — the tree ensembles — it is the better-calibrated choice, and it is the one that post-hoc recalibration can still improve. Two things would sharpen the picture: a held-out calibration split, which would remove the in-sample advantage the tree ensembles currently enjoy, and a finite-shot evaluation, which would establish whether any of this survives the noise of a real device. Neither is expensive, and the second is the question our simulation is constitutionally unable to answer.

AUTHOR CONTRIBUTIONS

Quan Tran Anh Vo: conceptualization; methodology; software; formal analysis; validation; visualization; writing – original draft; funding acquisition. **Quang Anh Nguyen:** software; investigation; data curation; formal analysis; validation; visualization. **Nang Hung Van Nguyen:** conceptualization; supervision; writing – review and editing. **Minh Tuan Pham:** supervision; resources; writing – review and editing.

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DATA AVAILABILITY STATEMENT

Every number reported in this paper is computed from artifacts archived with it. The stored per-run measurements,

the scripts that generate every figure and table, and the audit scripts that recompute each reported number from those artifacts are openly available at <https://doi.org/10.5281/zenodo.22893683>. Running `audit_p2_rebuild.py` from that archive reproduces every number this paper prints and fails if any disagrees. The two corpora are public benchmarks: NSL-KDD [1] and UNSW-NB15 [30].

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

This study used only public benchmark datasets of network traffic records and involved no human participants, no animal subjects and no personally identifiable information. Ethical approval was therefore not required.

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